

Physics-Informed Surrogate Modeling for Ra-226 to Ac-225 Transmutation

A computational study of whether a recurrent neural model can reproduce a reduced five-nuclide transmutation system more accurately than a matched non-recurrent control. The long-term goal is faster, auditable screening for Ac-225 production research.

V3 SEED-42 RUN COMPLETE

COMPUTATIONAL PROOF OF CONCEPT

EXTERNAL VALIDATION NEEDED

Sam Ogunnubi

12th-grade student researcher
Glen Burnie High School | AACPS | Maryland, USA
Computational nuclear engineering and machine learning

CURRENT EVIDENCE-BASED CONCLUSION

On a frozen synthetic benchmark, the seed-42 V3 LSTM achieved a 0.97% worst-mesh median Ac-225 endpoint error and outperformed a parameter-matched MLP on the predeclared paired median criterion. A hidden test using intentionally different conditions reached a 1.56% median point estimate, but its uncertainty interval did not establish a universal below-3% guarantee.

0.97% Matched LSTM median 111 positive Ac-225 cases; worst of four mesh medians	2.91% Matched MLP median Same seed, data contract, inputs, schedule, and tests
-0.63 pp Paired median difference 95% CI: -1.09 to -0.30 percentage points; favors LSTM	1.56% Hidden shifted-test median Different conditions from training; 95% CI: 0.69% to 7.60%, so this is not a guarantee

Five-nuclide reduced system | 1,400 synthetic training cases | 6,000 epochs | seed 42 | meshes 32, 48, 64, 96 | NVIDIA A100

What I Built and Verified

Evaluated nuclear inputs	Exact and Radau trajectories	Matched LSTM and MLP	Frozen tests and audit
Physics model: a reduced Ra-226 / Ra-227 / Ac-227 / Ra-225 / Ac-225 chain with spectrum-aware reaction inputs and Bateman dynamics. V3 architecture: an LSTM sequence surrogate constrained by initial conditions, endpoint objectives, atom-budget behavior, and physics residuals. Controlled comparison: 818,442 LSTM parameters versus 797,706 MLP parameters, a 2.53% budget difference, with verified paired scenarios and provenance. Engineering audit: 16 documented defects were repaired or isolated, including Jacobian, shielding, loss scaling, timing-input, random-stream, reporting, and Colab-resume failures. Checkpoint replay: local evaluation reproduced the cloud results; exact Bateman propagation and FP64 Radau agreed near numerical precision for this reduced system.			

Matched Architecture Result

V3 LSTM		0.97%
Matched MLP		2.91%

Why this comparison matters: V2 used a different scenario draw and scoring pipeline. The matched MLP is the valid control for testing whether V3's recurrent structure helped under this protocol.

SUPPORTED BY THE COMPLETED EXPERIMENT

For seed 42 on this reduced synthetic benchmark, the recurrent V3 model met the 1-3% median point-estimate target on both untouched evaluations and beat the matched pointwise control on the declared paired median criterion.

NOT YET SUPPORTED

Universal 1-3% prediction accuracy, real reactor or isotope-production accuracy, general LSTM superiority across seeds, clinical validity, or superiority to exact Bateman, ORIGIN, or OpenMC calculations.

The Median Is Not the Whole Story

Distribution of each positive scenario's worst error across four evaluation meshes:



Open reliability problem: matched p95 was 99.6%, locked p95 was 31.2%, and fast-spectrum cases dominated locked failures. Rare exact-zero matched cases also produced false positives as high as 8.09×10^{16} atoms. These failures remain visible in the report.

Where External Collaboration Would Help Most

Highest-value contribution: an independent ORIGIN, OpenMC/depletion, measured irradiation, or production dataset that can test transfer beyond the project's own synthetic generator.

Additional help is welcome with reaction-channel review, uncertainty propagation, fast-spectrum failure analysis, and a confirmatory multi-seed design. The present reduced exact solver is already fast; the next scientific gain is stronger realism and validation, not another claim based only on a lower training loss.

Next Research Milestones

- 1

Freeze V3 and do not tune on either revealed test set.
- 2

Obtain one independent high-fidelity or measured benchmark.
- 3

Investigate fast-spectrum tails and zero-production failures.
- 4

Run additional seeds only if claiming general recurrent superiority.
- 5

Document uncertainty, provenance, and all negative results.
- 6

Present V3 as a computational proof of concept until externally validated.

Metric contract: relative error = $| \text{prediction} - \text{truth} | / \max(| \text{truth} |, 10^{12} \text{ atoms})$; positive-case reporting mask = 10^{10} atoms. Matched evaluation contained 220 total scenarios (111 scored positive cases); shifted locked evaluation contained 60 total scenarios (28 scored positive cases). Results are from one training seed.